

Is Public Transportation Usage a Function of Group Communication Apprehension?

Sasha Sweat, Brenna Pratt, Nick Mergendahl

today

Table of contents

1	Introduction	1
1.1	Lit Review	2
1.2	Research Questions	2
2	Data & Methods	3
2.1	Data Preparation	3
2.1.1	Data Collection and Merging	3
2.1.2	Data Cleaning and Coding	3
3	Results	3
3.1	RQ1: Does group communication apprehension correlate with public transportation usage?	3
3.2	RQ2: Does a person’s number of days they used public transportation per month on average over the last 3 months predict their composite communication apprehension score from responses to questions about their confidence in conversations?	4
3.3	RQ3: Do individuals who feel comfortable in group discussions use public transportation more on a regular basis?	5
4	Discussion & Suggestions for Next Steps	7

1 Introduction

This manuscript is our submission for the summer semester project for Psych 755 at UW-Madison and its course entitled Tools for Large Scale Data Science. In this project we

demonstrate usage of the following tools: Python, R, Markdown, Quarto, Github, and Agentic IDEs.

1.1 Lit Review

There is existing research that indicates a significant negative correlation between high group Communication Apprehension (CA) scores and public transportation frequency usage. Because public transit settings have been indicated to predict heightened levels of generalized anxiety and distress among adult populations (@SarmientoSuarez2026), it is reasonable that individuals with elevated baseline communication apprehension would avoid high-stimulus, socially dense environments, like public transportation, disproportionately when compared to more relaxed peers.

While there is substantial evidence that access to robust public transportation can actually improve mental wellbeing (@cdcfoundation2020hi5), there is also substantial research on transit-induced stress that may support our hypothesis that public transportation usage is a function of group Communication Apprehension.

1.2 Research Questions

Our overarching research question explored whether public transportation usage was a function of group communication apprehension. We investigated that relationship with the following questions:

- **RQ1:** Does group communication apprehension correlate with public transportation usage?
 - H_0 : There is no correlation between group communication apprehension and public transportation usage.
 - H_a : There is a correlation between group communication apprehension and public transportation usage.
- **RQ2:** Does a person's number of days they used public transportation per month on average over the last 3 months predict their composite communication apprehension score from responses to questions about their confidence in conversations?
 - H_0 : Individuals who feel comfortable in group discussions do not use public transportation more on a regular basis.
 - H_a : Individuals who feel comfortable in group discussions do use public transportation more on a regular basis.
- **RQ3:** Do individuals who feel comfortable in group discussions use public transportation more on a regular basis?

- H_0 : Individuals who feel comfortable in group discussions do not use public transportation more on a regular basis.
- H_a : Individuals who feel comfortable in group discussions do use public transportation more on a regular basis.

2 Data & Methods

2.1 Data Preparation

2.1.1 Data Collection and Merging

Our data was collected from Prolific and Qualtrics Survey responses. We combined two Prolific demographic datasets (File A and File B) using pandas concatenation, then merged them with the Qualtrics survey responses. The merging process involved standardizing participant ID columns across datasets and performing an inner join on participant IDs to ensure that only complete records would be retained in the final merged dataset.

2.1.2 Data Cleaning and Coding

We prepared our data for analysis by recoding survey responses to questions about communication apprehension to numeric values on a 5-point scale: Responses of “Strongly disagree” to questions about communication apprehension were coded as 1, “Somewhat disagree” as 2, “Neither agree nor disagree” as 3, “Somewhat agree” as 4, and “Strongly agree” as 5, and the reverse was true for reverse coded questions that asked about communication confidence. We then added the total number for each question response about communication apprehension or confidence together to get a total composite score for communication apprehension.

This table explains the key descriptive statistics of our merged dataset. The communication apprehension mean score represents the average response across 12 survey items (Q1-Q6, Q13-Q18) measuring comfort with group communication, with appropriate reverse coding applied to items Q2, Q4, Q6, Q14, Q16, and Q17. The public transportation usage mean score represents the average frequency of public transportation use across 4 survey items (Q26-Q29).

3 Results

3.1 RQ1: Does group communication apprehension correlate with public transportation usage?

To address RQ1, we conducted a simple linear regression analysis examining the correlation between group communication apprehension and public transportation usage. The analysis used

the mean communication apprehension score (calculated from 12 survey items with appropriate reverse coding) as the predictor variable and mean public transportation usage score (calculated from 4 survey items) as the outcome variable.

Table 1: Regression Results

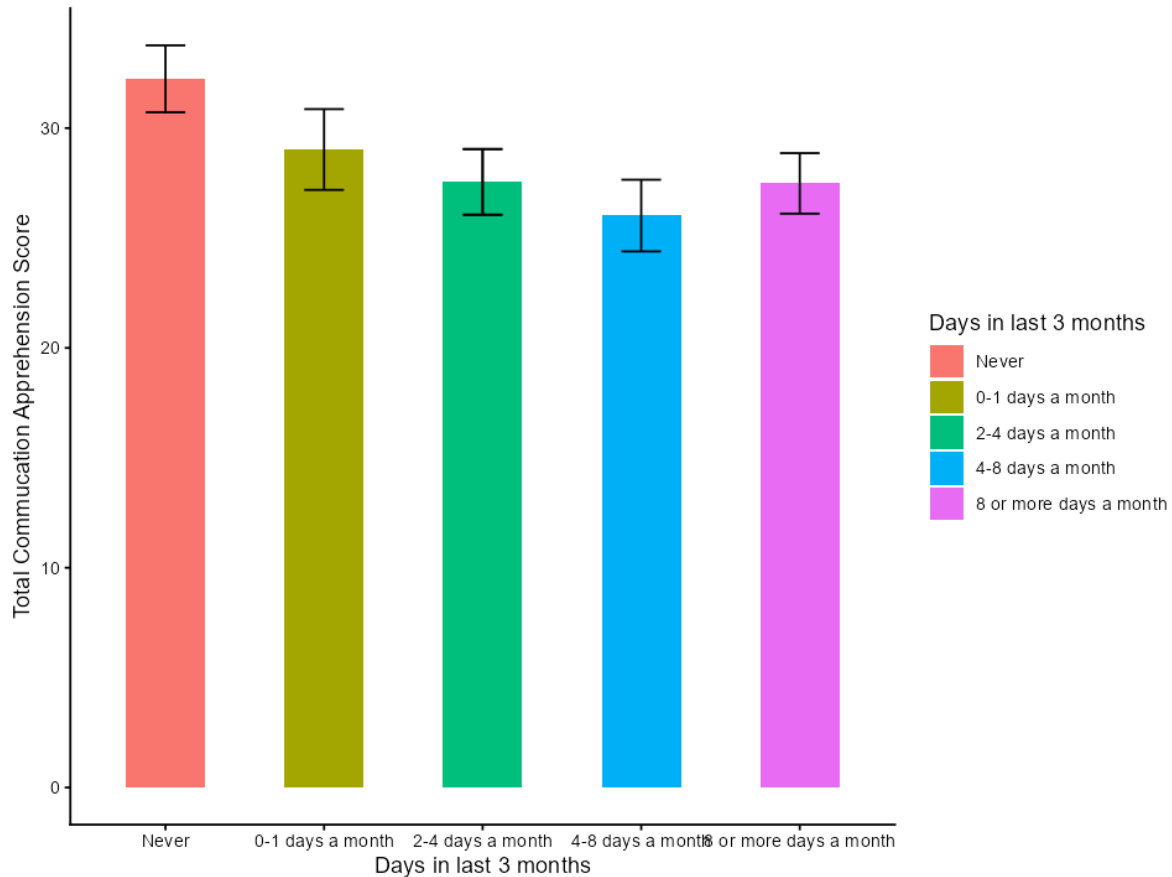
	R-squared	Sample size	Slope	Intercept	p-value	Standard error
0	0.108	252	-0.4147	2.8160	0.0000	0.0753

Figure 1: Regression Results

The regression analysis revealed a significant negative relationship between group communication apprehension and public transportation usage. The slope of -0.4147 indicates that as communication apprehension increases, public transportation usage decreases. The R-squared value of 0.108 implies that approximately 10.8% of the variance in public transportation usage can be explained by communication apprehension. The p-value < 0.001 indicates that this relationship is statistically significant, which led us to reject the null hypothesis and conclude that there is a correlation between group communication apprehension and public transportation usage.

3.2 RQ2: Does a person's number of days they used public transportation per month on average over the last 3 months predict their composite communication apprehension score from responses to questions about their confidence in conversations?

Our research memo for Student C discussed an ANOVA model determining whether a participant's response to a survey question asking how many days they use public transportation in a month on average could predict their calculated composite score for communication apprehension. The test showed that the response could predict the composite score, with the mean composite communication apprehension scores of respondents who said they never used public transportation shown to be significantly higher than those who used public transportation more than 7 days in a month.



This chart represents the average composite communication apprehension score across all participants based on their answer to the question about the number of days they used public transportation in a month, with margins for significant error included. There is a significant difference seen between the mean composite communication scores for participants who answered that they never used public transportation and those who answered that they used public transportation more than 7 days in a month.

3.3 RQ3: Do individuals who feel comfortable in group discussions use public transportation more on a regular basis?

Figure 2 presents a violin plot visualization displaying public transportation use and comfort in group conversations: On the y-axis (ranging from never to 8 or more days a month) is a scale of how often participants have used public transportation in the last three months. The x-axis displays a Likert scale (ranging from strongly agree to strongly disagree) measuring how much the participants agreed with the statement “Generally, I am comfortable while participating in group discussions”. We are shown narrowing violin shapes clustered around 8 or more days a month for individuals that strongly and somewhat disagreed with the statement.

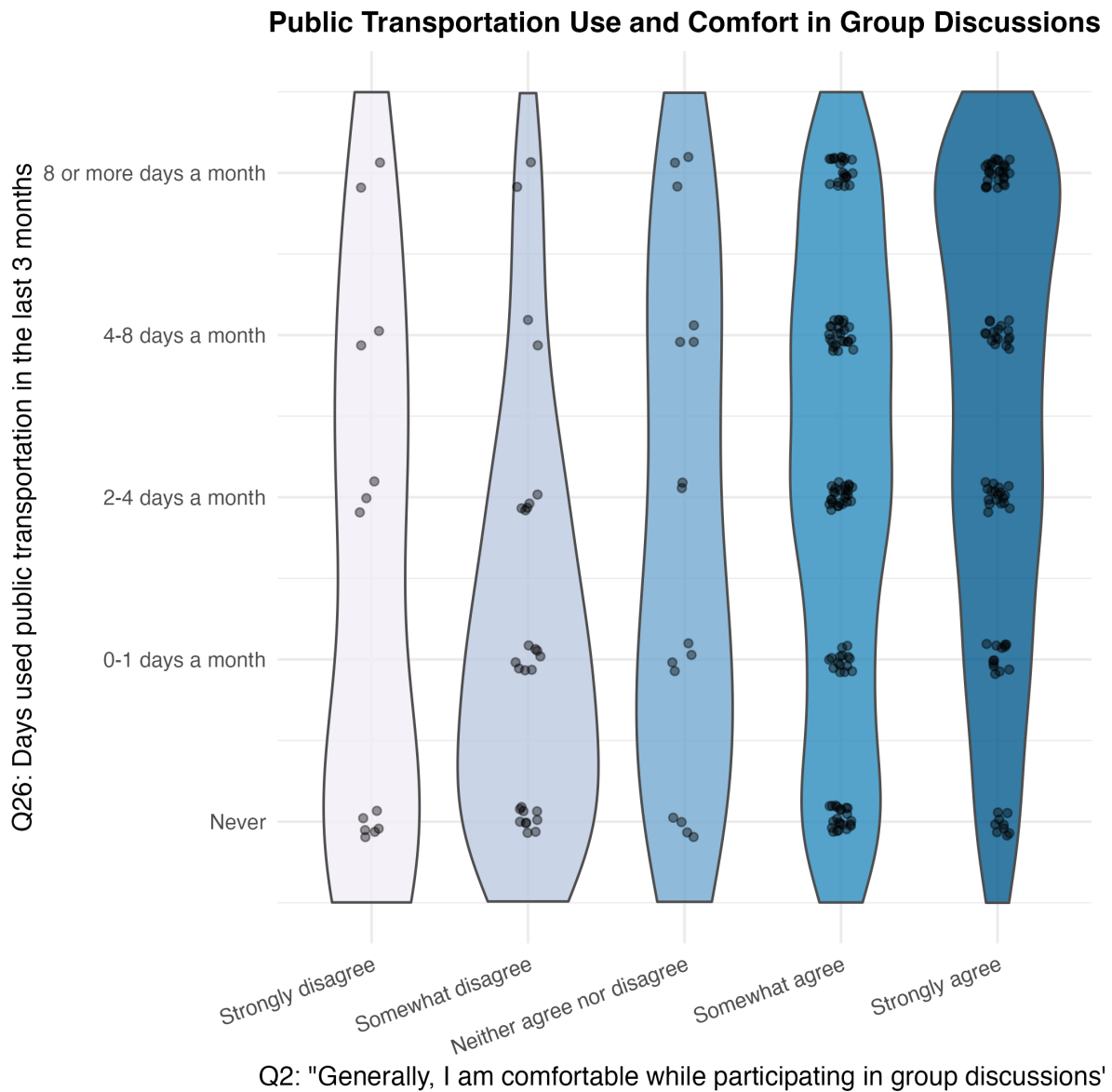


Figure 2: Violin Plot Comparing Public Transport Use (Q26) and Comfort in Group Discussions (Q2)

The violin plot also displays much wider distributions extending around never. Together, these visuals demonstrate that individuals who don't feel comfortable in group discussions use public transportation less in a given three month period.

The violin plot illustrates that individuals who answered "strongly agree" to the statement "Generally, I am comfortable in group discussions" reported using transportation 8 or more days in a month at a higher rate than those who answered "strongly disagree". Individuals who answered strongly disagree also reported never using public transportation in the last three months at a much higher rate than those who answered "strongly agree".

4 Discussion & Suggestions for Next Steps

This analysis establishes correlation but cannot determine causality. Nature vs. Nurture: Does higher communication apprehension actually lead to reduced public transportation use, or do individuals with different transportation patterns develop different communication styles? The communication apprehension scale needs validation to ensure it reliably measures interest or comfortability in group settings. Additional control variables (urban/rural, income, accessibility needs, prevalence of transportation network) may influence an individual's communication patterns and transportation choices. The sample also may have selection bias from Prolific participants, which limits generalizability. It seems clear that the number of days a participant has used public transportation per month over the last 3 months reliably predicts their composite communication apprehension score. However, it is not yet clear how other variables can influence these predictions. How can variables like age, gender, and employment status affect these results?